

Ananmay Som Singh

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EDUCATION

University of Maryland, College Park | College Park, MD

Expected May 2028

Bachelor of Science in Computer Engineering

Relevant Coursework: Algorithms, Data Structures, Computer Systems, Compilers, Probability & Statistics

EXPERIENCE

GSAAlpha Labs | Real Estate Transaction Automation

May 2026 – Present

Software Engineering Intern · Part-time during academic year

San Francisco, CA

- Built HomeFlow AI, a Next.js/TypeScript platform that files closing packets, drafts offers, and tracks deadlines for California agents; in production, entering its first agent pilot.
- Cut purchase-agreement data extraction from 35 minutes of manual entry to under 30 seconds with an LLM pipeline that reached 96% field accuracy on 15 real transactions; a human reviews before filing.
- Shipped the offer workflow that turns a 175-page disclosure packet into a drafted purchase agreement in under 5 minutes, sending only the conflicting fields to the agent for a decision.
- Improved document classification from 14/27 to 27/27 correct on the first real closing packet by matching form-specific footers instead of in-text references to other forms.
- Automated contingency deadlines across California business days and holidays, displaying the rule behind each date in response to agent distrust of auto-filled dates, with every change logged for disputes.

theconviction.ai | Financial Research Automation

May 2025 – Aug 2025

Software Engineering Intern

Remote

- Replaced 12 hours a week of manual analyst data collection with a FastAPI/PostgreSQL pipeline that pulls filings, transcripts, and news for 50+ companies into one record every morning.
- Added timeouts, retries, and deduplication to handle unreliable sources and overlapping feeds, keeping one clean record per company.
- Shipped a Next.js research tool the 4-person analyst team used daily; it puts each company on a timeline and links every finding to its source filing, verifiable in one click.

SEDS at UMD | SatFab CubeSat GPS Team

Sep 2024 – Feb 2026

Software Engineer

College Park, MD

- Built the Python test framework for 26 Verilog modules in a CubeSat GPS receiver and trained 6 engineers on it; it became the team standard for testing every new module.
- Cut the hardware regression from 4 hours to 95 minutes by running testbenches in parallel on a shared Linux server, so it runs before every merge instead of overnight.
- Automated waveform checking and pass/fail reporting so integration bugs were caught in simulation instead of on the hardware.

PROJECTS

Tape: Market Data Capture & Replay | *Go, AWS S3, ECS, CloudWatch, Terraform, Docker*

AnanmayS/tape

- Built a Go service that records live exchange feeds to S3 and replays them byte-for-byte, enabling reproducible market-data replay; sustained 49.8k messages/sec in load tests against a 37/s live feed.
- Detected all 3 feed gaps (3,240 lost messages) in fault-injection tests and flagged the affected windows before any backtest runs on them.
- Reduced feed-storage size by a factor of 5.2 using delta-encoded columnar storage.

ForgeGrid: Distributed Build System | *Node.js, JavaScript, Docker, Linux*

AnanmayS/forgelgrid

- Designed a Node.js build system that distributes tasks across worker machines as their dependencies finish; a cold build dropped 59% (2.5 s to 1.0 s) going from one worker to three.
- Added a content-addressed cache that skips any task whose inputs have not changed; an identical second build reused all 7 of 7 tasks.
- Handled worker crashes by reassigning their tasks, verified by tests that kill workers mid-build and check the output is still correct.

ShowdownRL: Reinforcement Learning Agent | *Python, PyTorch, PPO, Playwright, Gymnasium*

AnanmayS/ShowdownRL

- Trained a PPO agent on a 106-feature game view for Pokémon Showdown that won 79% of 1,000 simulated Random Battles against a type-aware scripted opponent; masked illegal moves so no turn is wasted.
- Drove real matches through the browser with Playwright and saved every battle log, so any reported win rate traces back to the games behind it.

TECHNICAL SKILLS

Languages: Go, Python, Java, C++, C, TypeScript, JavaScript, SQL, Verilog

Backend & Data: FastAPI, Node.js, PostgreSQL, Drizzle, SQLAlchemy, Supabase, Zod, WebSockets, Concurrency

Infrastructure & Tooling: AWS, S3, ECS, CloudWatch, Terraform, Docker, Linux, GitHub Actions (CI/CD), Vercel, pytest, Vitest

Frameworks & ML: React, Next.js, PyTorch, LLM extraction and evaluation